

Institute of Actuaries of India

Subject CP3 – Communication Practice

May 2023 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:

Thank you for your email. I have specific answers for your questions below, but I would like to assure you that we are monitoring the current situation of rising interest rates and our ability to manage the situation and write business as per our plans.

Your question is whether we have similar risks as banks and whether we would suffer in a situation in which a large number of customers decided to surrender. The answer is yes. You also wanted to know whether we would have to slow down on our plans. However, as I explain below our risk management framework has been designed to handle fairly extreme situations and as of now there is no threat to business.

We have two key risks, interest rate risk and liquidity risk. Interest rate risk arises because we commit to the customer, annuity rates based on our expected interest earnings. We therefore have to lock into interest rates for the amounts that we will receive in future. We do this by entering into derivatives which enables us to buy bonds at a later date with known yields. We also ensure that the sensitivity of the value of the assets and liabilities to movements in interest rates are matched through a duration matching strategy. What this means is that while interest rates increase, the value of the bonds that we hold goes down, but the value of our liability, that is the money that we owe the customers can be restated downwards. The movements in the value of the assets and liability are approximately the same and therefore the balance sheet is protected.

While entering into derivatives addresses our interest rate risk it also introduces us to another risk, liquidity risk. As and when interest rates change the value of the derivatives also change. If interest rates were to increase from the levels that we had bought the derivatives the value of the derivatives will go down. Our losses are gains for our counterparties and we need to pay them these amounts. In addition, liquidity risk arises because customers can surrender their policies and if there are large surrenders, then we would have to liquidate our assets.

Normally we expect a small percentage of our customers to surrender their policies for various reasons. We do levy a penalty so that we do not suffer a loss by this act of the surrender. It also protects us from the situation of customers selectively choosing to surrender when interest rates go up from the level which they have bought.

In order to meet these payouts as well as be able to pay our counterparties we invest a certain proportion of our assets in cash. We also monitor the liquidity situation by doing a stress test to see that our assets are sufficiently liquid to pay off our customers and our counterparties in an extreme situation like a high interest rate coupled with high surrenders.

While we will be able to meet these payouts we want to avoid a situation of having to suffer losses. In case the movement in interest rates and consequentially surrenders are high, we may need to sell assets at lower values which will result in losses and affect the capital position of the Company.

We conducted some additional stress tests to see what could be the impacts in more extreme circumstances. We are chiefly concerned about solvency and our ability to carry on with the business plan. Solvency should be healthy even if we write business as per plan.

We checked the impact on solvency of interest rates going up by 200 bps and simultaneously if all those customers who would benefit by surrendering their policies and buying fresh policies actually resort to doing this. We see that these are extreme scenarios and the solvency level is affected quite significantly. While the Regulatory limit is 100%, we have internally defined a threshold of 120% in a stressed situation which would require the Company to work towards improving the capital position. We see that in this situation the capital position of the Company goes to 120% as compared to the current level of 150%. Therefore, the Company has decided

on some interim measures like increasing its cash holding so as to avoid forced sale of assets and lower the amount of derivatives it enters into as there could be surrenders if interest rates keep rising. Of course, the Company will continue to monitor the situation. However as of now we can continue with our business plans.

I hope I have managed to give you an overview of our risk management strategy and answered your questions. Please do let me know if there are any further questions.

Regards

Risk Actuary

[90 Marks]

Solution 2:

Any answer which logically lays out the thought which has gone into drafting of the response covering the three key points.

E.g

I thought as head of sales the question that would be on top of his mind is whether there would be a change of business plan. Considering that there is a strategy meeting which is coming up where the current economic environment and response of the Company will be discussed he would want to be prepared to understand the risk management framework of the Company. He has also read various articles of banks going out of business and would want to understand whether the situation would be similar for an insurance Company. I took my cue from his email and structured the answer in such a way that his key questions are addressed.

On depth of the answer, I gave him information which he has specifically requested. While he is part of the management and would be involved in decision making, it is not necessary for him to know the specifics of matters. In making its recommendation, the risk management Committee would have recourse to the policy which has already been drawn up and would be assessing particularly any risk which is arising from the current situation which the current framework may not address sufficiently. Considering that a strategy meeting is planned, the sales director should have enough knowledge on the framework to participate in any discussions. He is also asked specific questions after reading about the situation in other countries.

I thought it is best to explain how the risk management framework works, what is the level of stress that the Company can handle given its business plans. I thought it is important he understands what are the key risks faced by the Company and therefore explained it in as simple a language as possible. I also explained that liquidity risk is something that the Company keeps monitoring that the current framework allows for extreme situations. I did not think it was really necessary that he knows the precise results as they would not have really increased his understanding of the situation. I did emphasise on the conclusions. I also pointed out that while risk management has factored in stresses, considering the current situation, stresses have been performed on top of the current stressed situation. Tactical measures have been put in place to handle the situation, but the Company is capitalised to handle these situations.

I didn't think there was a need to point how pricing was being done, or the kind of asset strategy which was being followed and definitions and ratios of liquidity coverage ratio.

[10 Marks]
